Durable Alignment via Orthogonal Memory in Large Language Models

A bounded truth-maintenance substrate over a frozen base model

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Abstract

The model is frozen. The world it serves is not. Corrections installed today need to land on the right region, revise when the facts change, and still be there after the base model is fine-tuned again. None of the standard tools does all three.

Put the corrections outside the model. A separate field — gradient-isolated from the weights — carries them, and its dynamics are the dynamics of the Informational Buildup Framework [1]: particles nucleate where the base model is wrong, crystallize where they keep being right, and dissolve where later experience contradicts them.

On a frozen Mistral-7B and a synthetic company of 1,000 employees, the field installs facts the base model never saw, overrides counterfactuals at 88.7% accuracy, deletes one employee's [DEL_N_FACTS] facts with near-zero drift on matched neighbors, and survives LoRA, instruction tuning, and twenty rounds of adversarial inject–retract with bounded amplitude.

Alignment here is *persistence*, not specification: the mechanism holds a target the designer chose. And it holds what is specified or compiled, not what it would have to derive on its own.

Keywords: durable alignment; substrate decoupling; truth maintenance; frozen LLMs; kernel memory; lifecycle operations; continual learning.

1 The Durability Problem

Train a language model. Deploy it inside a company. By Monday the engineering team has a new initiative, the data-org manager has changed, one employee has left. The model is wrong about all three. It will keep being wrong until someone teaches it otherwise.

The three standard answers each break differently. *Fine-tuning* adjusts shared parameters; the new facts land and old ones are perturbed at the same time [4, 5], because knowledge is not stored in discrete units [2, 3]. *Direct editing* (ROME, MEMIT) targets the perturbation [6, 7], and the perturbation works the day it is applied; the next round of fine-tuning often erases it. *Retrieval* [8] sidesteps the model entirely — the fact lives in an index, and the model is shown it at inference — so behavior changes only when retrieval fires *and* the prompt rewards the retrieved fragment over the prior. Neither is guaranteed.

The shared failure: behavior installed today does not reliably remain installed tomorrow, in the right region, after the world has moved and the model has been touched again. Call that property *durable alignment*. The word *alignment* is doing narrow work — the operator chose the target elsewhere; the question is whether the system holds it.

This paper builds a mechanism that does. The corrections live outside the model, in a substrate the base-model gradient cannot reach, evolving under the dynamics of the Informational Buildup Framework [1].

Running example. *Future Industries*: a synthetic company of 1,000 employees, each with structured facts — manager, team, project, mentor, location — that move through onboarding, a new initiative, a reorganization, and turnover. Facts are added, extended, revised, retired. The model is never told which is which. Small enough to instrument, large enough to break a naive mechanism.

2 An Orthogonal Memory

Do not edit the model. Sit a second memory on top of it.

The base model produces a coherence score $\widehat{\mathcal{R}}_{\text{base}}(x, y)$ over candidate outputs. The memory adds a local correction $\delta\widehat{\mathcal{R}}(x, y)$:

$$\widehat{\mathcal{R}}^{\text{eff}}(x, y) \;=\; \widehat{\mathcal{R}}_{\text{base}}(x, y) \;+\; \delta\widehat{\mathcal{R}}(x, y). \tag{1}$$

Two facts about Eq. (1) carry the rest of the paper.

No gradient path. $\delta\widehat{\mathcal{R}}$ is parameterized independently of $\widehat{\mathcal{R}}_{\text{base}}$. Updating one leaves the other alone. Whatever the base model becomes tomorrow, yesterday's corrections are still there.

Locality. $\delta\widehat{\mathcal{R}}$ rises and falls in regions of representation space, not across the model. A correction at one point does not leak across all decisions.

Intuitively, $\widehat{\mathcal{R}}_{\text{base}}$ is what the model believes; $\delta\widehat{\mathcal{R}}$ is a separate file of corrections written in a language the model cannot edit. Decisions read both.

2.1 What the memory is

The same engine that ran the theory paper [1], now sitting on top of a frozen LLM. $\widehat{\mathcal{R}}_{\text{base}}$ is the baseline coherence evaluator; $\delta\widehat{\mathcal{R}}$ is the experience-modified component of $\widehat{\mathcal{R}}^{\text{eff}}$. The engine writes corrections as kernel-localized particles in latent space and lets them live, change, or die under a single dynamical rule.

With $z = \phi(x, y)$ the representation of a candidate pair,

$$\delta\widehat{\mathcal{R}}(z) \;=\; \sum_i \gamma_i\, v_i\, \mathcal{K}(z, z_i), \tag{2}$$
$$\mathcal{K}(z, z_i) \;=\; \exp\Big(-\tfrac{\|z-z_i\|^2}{2\sigma_i^2}\Big).$$

Each particle has a position z_i, an amplitude v_i, a width σ_i, and a read-gating flag $\gamma_i \in \{0, 1\}$ that decides whether it speaks in the current context.

The driving signal is the gap between the environment's coherence and the system's:

$$\mathcal{D}(z) \;=\; \widehat{\mathcal{R}}^{\text{ext}}(z) \;-\; \widehat{\mathcal{R}}^{\text{eff}}(z), \tag{3}$$

with $\widehat{\mathcal{R}}^{\text{ext}}$ the supervised target: the operator says what should be coherent.

The field then evolves under the Modification Postulate of the theory paper:

$$\frac{\partial}{\partial t}\delta\widehat{\mathcal{R}}(y) \;=\; \eta\, \mathcal{K}(y, z_S)\, \mathcal{D}(z_S) \;-\; \mu_{\text{eff}}\, \delta\widehat{\mathcal{R}}(y). \tag{4}$$

2.2 What the memory does

That one equation is the whole lifecycle.

A particle *nucleates* where the discrepancy is large and no existing particle covers the region. It *reinforces* where the same correction keeps being right, accumulating amplitude. When repeated reinforcement converges, the particle *crystallizes*: its decay rate drops from μ_{base} to μ_{cryst} and it becomes long-lived structure. If later cross-context exposure systematically contradicts it, the Crucible fires: decay returns to μ_{base}, broadcast rights are revoked, and the particle *dissolves*.

Intuitively, the field grows where it is wrong, hardens where it is right, and melts where it is later proved wrong again. Nothing else is added by hand.

This is what makes deletion clean. A contradicted fact reverses the discrepancy on the exact particles that carried it; they lose support and dissolve, neighbors do not (§4: drift on the deleted entity, near-zero drift on matched controls).

And it is what makes durability free. Fine-tuning rewrites $\widehat{\mathcal{R}}_{\text{base}}$ and changes nothing in $\delta\widehat{\mathcal{R}}$ (§6).

3 Experimental Setup

Future Industries. 1,000 employees, each with structured attributes (manager, team, project, mentor, location), moving through four phases:

- **A: Onboarding** — initial facts across all categories.
- **B: Initiative** — two new categories (certification, committee).
- **C: Reorganization** — counterfactual updates (managers, projects).
- **D: Turnover** — partial replacement and reassignment.

Each phase introduces new information and contradicts old information. The system is not told which is which.

Task. Multiple-choice over four candidates, one correct. The decision rule is Eq. (1) taken to its arg max:

$$y^* = \arg\max_y \left(\widehat{\mathcal{R}}_{\text{base}}(x,y) + \delta\widehat{\mathcal{R}}(x,y)\right). \quad (5)$$

A local preference over candidates, not a global modification of the model.

Model. Mistral-7B, frozen throughout.

Representation. A frozen MPNet encoder maps each (x, y) pair to $\mathbb{R}^{80}$: 64 dims of PCA-reduced semantic embedding, 8 subject features, 8 answer features. The feature blocks give the memory room to separate entities that share an answer surface.

Training. Correct answer: $\widehat{\mathcal{R}}^{\text{ext}} = 0$. Incorrect answers: $\widehat{\mathcal{R}}^{\text{ext}} = 1$. The discrepancy in Eq. (3) drives the field through Eq. (4). Phases run in order, no replay buffer, no gradient into the base model.

Convergence. Per-phase training stops when the linear readout has settled: moving-average improvement under 0.003 over ten evaluation points, slope under 0.0001 per epoch, ≥ 100 epochs. Hard caps (A: 300; B/C/D: 200). The resulting artifact — the *canonical engine* — is what every later experiment uses.

What we measure. *Linear accuracy* from $\widehat{\mathcal{R}}^{\text{eff}}$ (what the memory does) and *log accuracy* from base logits alone (what the base would do without it). Margins are bucketed into *consistent*, *ambiguous*, and *incorrect* so confidence is visible alongside accuracy.

4 Canonical Lifecycle and Truth Maintenance

One field, four phases, in order. Where does it land?

4.1 Across A–D

Per-phase accuracy: A=[PHASE_A_ACC], B=[PHASE_B_ACC], C=[PHASE_C_ACC], D=[PHASE_D_ACC]. At the end the field carries [CENTERS_TOTAL] particles, [CENTERS_CRYST] crystallized and [DISSOLUTIONS_TOTAL] dissolved. Nothing bounds the count except the dynamics, and the dynamics suffice.

4.2 Provenance

If the field decides, the field is inspectable.

For a showcase employee, we decompose the score on each query into per-particle contributions: the top-10 particles account for [SHOWCASE_TOP10_CONC] of the signal, and [SHOWCASE_CRYST_VER_PCT]% of the contributors are crystallized and verified. Across the audit set, [AUDIT_CONSISTENT_PCT]% are consistent, [AUDIT_AMBIG_PCT]% ambiguous, [AUDIT_WRONG_PCT]% wrong. Every prediction traces to particles with known birth contexts.

4.3 Retraction

$A \to B$ becomes $A \to B'$. The retraction-target set is designated in advance (seed 2026).

Original target falls — [RETRACT_TORIG_B], [RETRACT_TORIG_C], [RETRACT_TORIG_D]. New target rises — [RETRACT_TNEW_B], [RETRACT_TNEW_C], [RETRACT_TNEW_D]. Nearest-neighbor controls drift by [NN_DRIFT], distant controls by [DISTANT_DRIFT]. The change is contained.

4.4 Selective deletion

Pick one employee, [DEL_EMP]. Remove all [DEL_N_FACTS] facts about them.

Target: [DEL_BEFORE] → [DEL_AFTER]. Everyone else: [DEL_OTHERS_BEFORE] → [DEL_OTHERS_AFTER], change [DEL_OTHERS_DIFF]. The Crucible removed [DEL_CENTERS] particles. Nothing else moved.

4.5 Backward transfer

In a model without the dynamics, forgetting is global. Here it should appear only where contradiction is.

$\mathrm{BT}_{A\to B}$=[BT_AB], $\mathrm{BT}_{A\to C}$=[BT_AC], $\mathrm{BT}_{A\to D}$=[BT_AD_TOTAL]. On the $B \to C$ slice, where Phase C contradicts a subset of Phase B, affected entities show [BT_BC_AFFECTED] and unaffected ones [BT_BC_UNAFFECTED] (selectivity [BT_BC_SELECTIVITY]). Forgetting tracks contradiction, not exposure.

5 Strong Priors, Locality, and Compiled Semantic Structure

5.1 Strong-prior override

Future Industries is fictional, so the base model's priors over it are weak. To force the harder regime, we install corrections that disagree with strong priors on real entities. Override rate: [OVERRIDE_PASS]. Bleed on matched neighbors stays within [LOCAL_BLEED_BUDGET]. The field overrides where it is told to, stays quiet where it is not.

5.2 Ontology shift

Reorganize the category structure — title hierarchies, group taxonomies — and the field absorbs the shift through the same lifecycle: contradicted particles dissolve, new ones nucleate, the rest are untouched. No global reset.

5.3 Locality, scale, and bleed

Installation density sets how cleanly corrections stay local. Too sparse, the kernel covers ground it should not; too dense, adjacent installs blur. The sweep quantifies the working range.

5.4 Compiled semantic structure

Three cells draw the boundary between *enforcement* and *derivation*.

Explicit. Install $A \to B$ and $B \to C$ directly: the field enforces both. One-hop and two-hop consequences land cleanly.

Closure (negative). Install only $A \to B$ and $B \to C$. Does $A \to C$ emerge from the dynamics alone? It does not. Transitive closure does not arrive for free: the local rule in Eq. (4) writes corrections at visited regions, not at logical consequences of visited regions.

Compiled. A deterministic external procedure derives $A \to C$ and feeds it back as a fact. The field enforces it as durably as any direct install. When the closure changes — $B \to D$ replaces $B \to C$ — the derived consequence updates the same way: old $A \to C$ retracts, new $A \to D$ installs, controls do not move.

Intuitively, the field is a durable enforcement layer over whatever logical structure the operator supplies. It does not derive on its own; given a derivation, it makes the result stick and revisable.

6 Durability Under Base-Model Evolution

Disjoint substrate is an architectural claim. The point of this section is the empirical one.

6.1 LoRA

Train a LoRA adapter on Mistral-7B, merge it, re-evaluate the unchanged field. Light: flip-rate [LORA_LIGHT_FLIP], Δ=[LORA_LIGHT_DELTA]. Moderate: [LORA_MOD_FLIP], Δ=[LORA_MOD_DELTA]. Worst-case across conditions: [MAX_LORA_DELTA].

6.2 Instruction tuning

A separate instruction-tune: [INSTR_FLIP], Δ=[INSTR_DELTA].

6.3 Adversarial fine-tuning

Try to break it on purpose. The adversarial setup saturates the base model against the installed corrections under [ADV_PARAMS]: flip-rate [ADV_FLIP], Δ=[ADV_DELTA], ceiling C=[CEILING_C]. The point at which the field breaks gives the bound, not just the working range.

6.4 Continuous inject–retract pressure

Twenty rounds of alternating install and retract, Crucible on vs off as control.

Crucible on: universal-fact amplitude peaks at [VMAX_U_FULL_PEAK] (round [VMAX_U_FULL_PEAK_ROUND]) and settles to [VMAX_U_FULL_FINAL]; adversarial amplitude peaks at [VMAX_A_FULL_PEAK], settles to [VMAX_A_FULL_FINAL]. First clamp at round [FIRST_CLAMP_ROUND].

Crucible off: [VMAX_U_NOC_FINAL] and [VMAX_A_NOC_FINAL] — without dissolution, amplitudes grow. Frozen-field control drifts by [FROZEN_DRIFT]. The dynamics are doing the work.

7 Baselines and Cross-Model Generality

ROME, MEMIT, SERAC, GRACE, WISE need a separate environment and are deferred (Limitation 4). What follows is kNN, RAG, and a different base model.

7.1 kNN

A straight kNN over the same representation. Phase accuracies: [KNN_A], [KNN_B], [KNN_C], [KNN_D]. [KNN_INTERPRETATION].

Under oracle memory surgery — every retraction and deletion manually applied to the index — kNN is hard to beat. That is the unstated cost. The contrast is not final accuracy; it is whether the lifecycle is native to the substrate or imposed from outside.

7.2 RAG

Standard retrieval-augmented over the same corpus: [RAG_A], [RAG_B], then [RAG_C_UPDATED] with index updated and [RAG_C_STALE] without, [RAG_D]. The Phase C split is the point: RAG handles Phase C if the index is maintained, and not otherwise. Maintenance is the unstated assumption.

7.3 Cross-model: Qwen2-1.5B

Swap Mistral-7B for Qwen2-1.5B. Train a *fresh* $\delta\widehat{\mathcal{R}}$ field over Qwen's $\widehat{\mathcal{R}}_{\text{base}}$. Phase accuracies: [QWEN_A], [QWEN_B], [QWEN_C], [QWEN_D].

This is generality at the mechanism level, not zero-shot field transfer. The same dynamics work on a smaller, architecturally different base; the learned field does not move with it (Limitation 3).

7.4 Failure modes

kNN fails when the memory is stale. RAG fails when retrieval misses. The IBF field fails most clearly under paraphrase drift (§8). The shapes differ.

8 Paraphrase Geometry

A correction installed at z covers a kernel-shaped neighborhood. Whether a paraphrase lands in that neighborhood is a property of the encoder, not the engine.

ZsRE paraphrases typically stay inside the neighborhood of the direct install: the correction transfers. CounterFact paraphrases often land outside it, especially when noisy context shifts the prompt representation: the correction does not transfer. The audit measures, for each install, how far matched paraphrases sit in z-space and whether the kernel covers them.

Intuitively, the field cannot enforce structure on points it cannot reach. The fix is in the representation, not the dynamics.

9 Claim-to-Cell Map

10 Limits and Scope

1. Enforcement, not derivation. The field enforces what is specified or compiled. Transitive closure does not emerge from local relation training alone (§5); when an external procedure derives it, the field holds it.

2. Paraphrase transfer is geometry-bound. ZsRE transfers, CounterFact often does not (§8). The fix is in the encoder, not the engine.

#	Claim	Supporting cells	Artifact	Status
1	Local durable alignment without weight editing	canonical lifecycle; benchmark IBF runner; LoRA durability	`canonical_*`	[STATUS_1]
2	Truth-maintenance lifecycle (install / revise / remove / rollback / retain)	A–D canonical; retraction; deletion; forgetting diagnostic; residue hygiene	`lifecycle_*`	[STATUS_2]
3	Override of strong local priors, preserved locality	strong-prior override; ontology shift; locality/bleed; scale	`priors_*`	[STATUS_3]
4	Compiled semantic consequences durably enforced and revised	explicit implications; closure diagnostic; compiled closure	`closure_*`	[STATUS_4]
5	Durability survives base-model evolution (LoRA / instruction / adversarial)	LoRA; instruction tune; adversarial saturation	`durability_*`	[STATUS_5]
6	Not reducible to kNN or RAG	kNN; RAG; comparison report; failure-mode analysis	`baselines_*`	[STATUS_6]
7	Cross-model generality at the mechanism level	Qwen2-1.5B fresh-field replication	`qwen_*`	[STATUS_7]
8	Paraphrase transfer is geometry-dependent	CounterFact paraphrase geometry audit	`paraphrase_*`	[STATUS_8]

Table 1: Claim-to-cell map. Artifacts live under `mmlu_ibf_out/` as JSON. Status values: `paper-grade`, `smoke-validated`, `diagnostic`, `deferred`.

3. Qwen is fresh-field, not zero-shot. Mechanism generality across base models, not transfer of the same learned field.

4. External-editor baselines deferred. kNN and RAG reported. ROME, MEMIT, SERAC, GRACE, WISE need a separate environment; an availability manifest is shipped, numbers are not.

5. Uniform-prior regime. Fictional entities give $\widehat{\mathcal{R}}_{\text{base}} \approx (0.25, 0.25, 0.25, 0.25)$. The $\delta\widehat{\mathcal{R}}$-only ablation (§4) reports [X] on Phase A and [Y] on Phase C with a uniform base. The strong-prior regime is untested.

6. Entity-feature augmentation. 8D subject and 8D answer blocks give entities near-unique signatures by construction. The stricter test — shared-answer-pool retraction with genuinely overlapping beliefs — would probe the Crucible without that scaffolding. The most important extension we flag.

7. Specification is out of scope. The mechanism is target-neutral. We contribute durability, not how the target is chosen.

8. Single agent. One field, one base model, one knowledge lifecycle.

9. Scale. 1,000 entities, 1,640 unique facts. The bound at 10^6 or 10^9 is not demonstrated.

11 Conclusion

Put the alignment somewhere the gradient can't reach, and it stays.

A synthetic company of 1,000 employees across four phases. Install, extend, revise, retract, delete, audit — one field, homogeneous operations. Durability measured under LoRA, instruction tuning, adversarial fine-tuning, and twenty rounds of inject–retract.

The engine is the same one validated in [1] on non-stationarity, chess, and continual image classification — no mechanism changes. What is new is the lifecycle on top, and the demonstration that the truth-maintenance operations of classical AI can be reconstructed as a continuous bounded substrate sitting above a language model rather than inside it.

All of it follows from one architectural fact: correction field and base-model parameters live in disjoint substrates, and the field's dynamics are the dynamics of the Informational Buildup Framework.

The alignment you install today is still there after tomorrow's fine-tuning. **That is the result.**

Reproducibility

One Jupyter notebook runs the full pipeline, base-model extraction through all validation experiments, and writes every reported number to JSON or pickle. End-to-end runtime is approximately 49 hours on a single A100/H100. Seeds are fixed independently per experiment: main pipeline SEED = 42, retraction-target designation seed = 2026, long-horizon split seed = 2027. Code and artifacts:

https://github.com/negulescu42/information-as-alignment

The companion notebook is `(IBF)Companion-LLM-Durable-Alignment.ipynb`; its reproducibility appendix records run mode, run id, git commit, branch, hardware, seeds, dataset files, major artifacts, and estimated runtimes.

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